MILITARY SPECIFICATION

ELECTRODES, CARBON, LIGHTING,

HIGH-INTENSITY SEARCHLIGHT SERVICE

1. SCOPE

1.1 Scope. This specification covers lighting electrodes for 24-inch high intensity carbon arc searchlights.

1.2 Classification. Electrodes shall be of the following types, as specified (see 6.1):

Type I - Positive electrodes.
Type II - Negative electrodes.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
TT-E-485 - Enamel, Semigloss, Rust-Inhibiting.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

STANDARD

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The material shall be as specified herein and shall meet the performance requirements specified herein.

3.2 Tolerances. Carbon electrodes having diameters not greater than 2 percent over nor less than 1-1/2 percent under that specified will be considered as meeting the dimension requirement for diameter. A variation of plus or minus 1/8 inch will be allowed in the length dimension of positive electrodes and 1/16 inch in length of negative electrodes.

3.3 Core sizes. The diameter of the negative core shall be approximately 0.1 inch. The diameter of the positive core shall be, in all cases, approximately one-half the outside diameter of the positive shell.

3.4 Type I - positive electrodes.

3.4.1 Shell. The positive electrode shall consist of a hard, plain, homogeneous, uncoated shell of nonflaming carbon in accordance with 3.12. The electrode shall be sufficiently straight over its entire length to pass through a tube 12 inches long, and have an internal diameter of 1/16 inch greater than the maximum allowed diameter of the positive shell. The shell shall be uniform in thickness for its entire length. One end shall be beveled at an angle of 30 degrees to the longitudinal surface of 1/8 inch.

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3.4.2 Core. The positive core shall consist of a hard-flaming material and shall run centrally in the shell for its entire length. It shall be of uniform diameter, homogeneous in texture, and shall contain no air holes or spaces, cross cracks, or injurious flaws, and shall be applied within the shell secure and hard enough to withstand all shocks to which the electrode may normally be subjected without crumbling or disintegrating.

3.5 Type II - negative electrodes.

3.5.1 Shell. The negative electrode shall consist of a hard, homogeneous shell of nonflaming carbon in accordance with 3.12. It shall be sufficiently straight to pass through a tube 5 inches long and have an internal diameter of 15/32 inch. One end of the electrode shall be beveled to a diameter of approximately 3/16 inch at the end, the bevel extending approximately 1 inch along the length of the electrode.

3.5.2 Core. The negative electrode shall be cored for the entire length with nonflaming material, applied in such a manner as to prevent crumbling or detaching from the shell.

3.6 Current rating. Electrodes shall be designed normally to operate at 75 amperes.

3.7 Dimensions. The dimensions of the positive and negative electrodes shall conform to table I.

Table I - Dimensions.

Positive shell		Negative shell	
Outside diameter	Length	Outside diameter	Length
Inch	Inches	Inch	Inches
0.433	18	0.433	8-3/4

3.8 Candlepower. When the electrodes are operated at 75 amperes in a 24-inch searchlight drum and 24-inch high-intensity searchlight lamp, air cooled (see 4.4.2), the candlepower shall be as shown in table II.

Table II - Candlepower.

Minimum value of total horizontal candlepower at 20 degrees from axis - not less than	Value of crater intensity of 20 degrees - not less than
Candlepower	Candlepower
35,000	13,000

3.9 Life per trim. When operated at 75 amperes the life per trim of a single pair of electrodes shall be 1-3/4 hours.

3.10 The electrodes shall pass the burning test specified in 4.4.1.

3.11 The electrodes shall be of such chemical composition and mechanical structure that when properly installed and adjusted in the searchlight, the arc color characteristics specified in 4.4.4 will be attained.

3.12 Workmanship.

3.12.1 Carbon electrodes shall be circular in cross section, uniform in diameter for the entire length, and free from all flaws, cracks, or impurities that may cause unsteady burning.

3.12.2 All cores shall be tight for the entire length within the shell, so that the electrode, when dropped on a plane surface, will give a true ring. Unless this true ring can be obtained, the workmanship will be considered unsatisfactory.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. All electrodes of the same type and polarity offered for delivery at one time shall be considered a lot for purposes of examination and tests.

4.2.2 Sampling for examination of filled outer shipping containers. A random sample of shipping containers filled with packages of electrodes shall be selected in accordance with MIL-STD-105 at inspection level I and Acceptable Quality Level = 2.5 percent defective to verify compliance with section 5.

4.2.3 Sampling for examination of filled packages. Upon satisfactory completion of examination of filled shipping containers, a representative sample of filled packages shall be selected in accordance with MIL-STD-105 at inspection level I and Acceptable Quality Level = 2.5 percent defective to verify compliance with section 5. The packages selected shall be taken from shipping containers selected in the container examination sample.

4.2.4 Sampling for examination of electrodes. A random sample of electrodes shall be selected in accordance with MIL-STD-105 at inspection level I and Acceptable Quality Level = 2.5 percent defective to verify conformance with this specfication not including tests. The electrodes selected in this sample shall be taken from packages selected in the packaging examination sample.

4.2.5 Sampling for tests. A random sample of electrodes shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-2, and shall be subjected to the tests specified in 4.4.

4.3 Examination.

4.3.1 Examination of packaging and packing. Each of the containers, both the outer shipping container and the filled packages, in a sample selected in accordance with 4.2.2 and 4.2.3 shall be examined for defects of construction, for unsatisfactory markings and for noncompliance with section 5. If the number of rejected items exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3.2 Examination of electrodes. Each of the sample electrodes selected in accordance with 4.2.4 shall be examined to determine conformance with 3.12. Any electrodes with one or more defects shall be rejected. If the number of rejected electrodes exceeds acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 Burning. The positive electrode, when burning under normal conditions, shall maintain a deep, well-formed crater, and the negative shall show a well-defined negative tongue emanating from its core for operating currents above 100 amperes. The arc formed on these electrodes, under normal conditions, shall be steady and shall not emit any excessive amount of soot or other substance which will seriously impair the reflective qualities of the reflector or the transmission qualities of the front door during a period of three complete trims of electrodes, with the ventilation of the drum normal. When burning under normal conditions, the electrodes shall burn with very little ash and shall not throw off chips.

4.4.2 Candlepower. The candlepower of the arc for the normal specified operating current and normal operation of the arc shall show a value of crater light and an intensity at the maximum point of the total horizontal candlepower distribution curve conforming to the values specified herein. These values shall be obtained from the electrodes after the pair has burned at least 10 minutes when tested in a searchlight of the type specified in 3.8.

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4.4.3 Life per trim. The life per trim of a single pair of electrodes shall be in accordance with 3.9. At the end of the prescribed time there shall be sufficient length for both positive and negative electrodes remaining so that the studs still are in electrical contact with conducting fingers, the positive crater shall be in focus, there shall be normal voltage across the arc, and the electrodes shall be still under the control of the feeding mechanisms.

4.4.4 Arc color characteristics. The color of a beam working under normal conditions shall appear bluish white to the eye and shall show such color characteristics and intensity that not less than 60 percent of the emitted light from a flaming arc shall be within a wave length band from 0.50 to 0.60 micron.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.2.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A. Electrodes shall be preserved-packaged in quantities of 25 to a package, positive and negative separate. Method of preservation shall be in accordance with method IC-4 or IC-5 of MIL-P-116. Alternatively, electrodes may be packaged in metal cans which will pass the waterproofness test of MIL-P-116. Cans shall be coated on the exterior with a rust-resistant coating conforming to TT-E-485. Unit containers shall be of a type which can be opened and resealed readily.

5.1.2 Level C. Preservation-packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use.

5.2 Packing.

5.2.1 Levels A and B. Electrodes packaged as specified shall be packed level A or B as specified (see 6.1), in accordance with the requirements of MIL-E-17555.

5.3 Marking. Shipment markings shall be in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and quantity required (see 1.2).
(c) Level of preservation-packaging, and packing required (see 5.1 and 5.2.1).

6.2 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.3 THE MARGINS OF THIS SPECIFICATION ARE MARKED "*" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 5977-N031)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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SUBMITTED BY *(Printed or typed name and address - Optional)*

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